MASTER MINDS

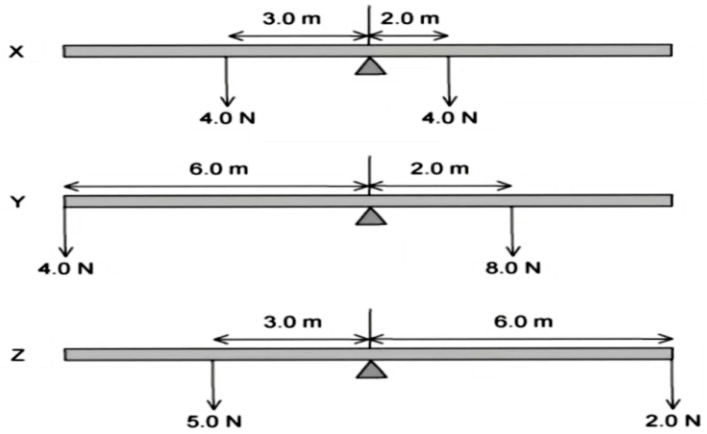
PHYSICS REVISION EXAM [FORCE] **Marks:-25**

SECTION A (Choose the correct answer)

1. Which of the following is true when a body is stationary? [5 marks]

   a. The force acting on it is not in contact with it
   b. The body is in vacuum
   c. There is no force acting on it
   d. The combination of forces acting on it balance each other

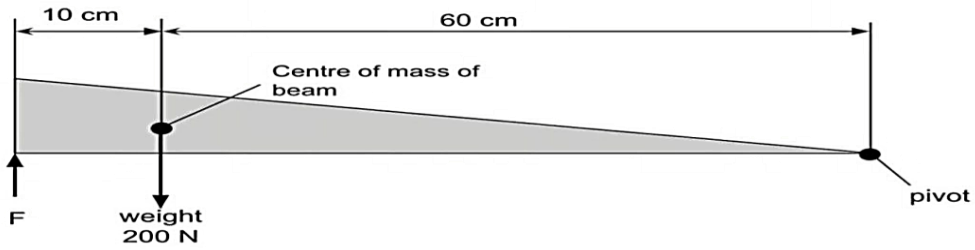
2. The diagrams below show three beams, each is pivoted at the centre. Each beam a has two forces acting on it, as shown.



   Which of the beam will rotate clockwise?

   a. X and Y
   b. Y and Z
   c. X and z
   d. None of these

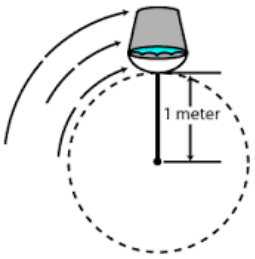
3. A non-uniform beam is attached to a wall by a pivot at one end as shown in the diagram below. The beam has a weight of 200 N.



   What is the approximate value of X

   a. 12000 N
   b. 171 N
   c. 120 N
   d. 12 N

4. You swing a bucket of water attached to a string in a circle above your head (refer give picture) What keeps the water in the bucket?

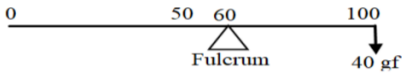


   a. Friction
   b. Centripetal force
   c. Inertia

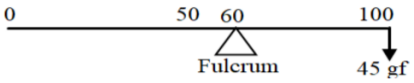
d. Gravity

5. If the centre of gravity of a metre scale of mass 80 g lies at the 45 cm mark, then which one of the following diagrams will show the balanced position of the scale?

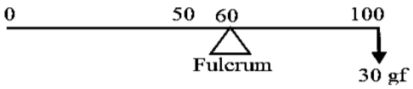
a.



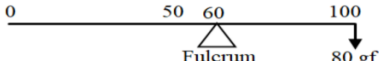
b.



c.



d.



SECTION B (attempt all)

1. Define moment of force and state its SI unit. [2 marks]
2. Screw jack is provided with a longer arm. Why? [2 marks]
3. State the principle of moments as applied to a beam in equilibrium. Name a device based on the above principle. [2 marks]
4. Differentiate between centripetal force and centrifugal force. [2 marks]
5. What is the position of centre of gravity of a triangular lamina? [2 marks]
   a. When this triangular lamina is suspended freely from any one vertex, what is the moment of force produced by its own weight in its rest position?
6. A uniform metre scale is balanced at 60 cm mark such that two weights one of 5 gf and another of 40 gf are suspended at 10 cm mark and 80 cm mark respectively. Calculate the weight of the metre scale. [2 marks]
7. A stone of mass m kg is rotated in a circular path with uniform speed by tying a strong string with the help of your hand. Answer the following questions:
   (i) Is the stone moving with uniform or variable speed?
   (ii) Is the stone moving with uniform acceleration?
   (iii) In which direction does the acceleration act?
   (iv) What kind of force acts on the hand? State its direction. [4 marks]
8. A boy of mass 30 kg is sitting at a distance of 2 m from the middle of the see-saw.Where should a boy of mass 40 kg sit so as to balance the see saw? [2 marks]
9. The diagram shows wheel O pivoted at point A. Three equal forces F1, F2 and F3 act at point B on the wheel.
   a. Which force will produce maximum moment about A? [2 marks]
   b. Give a reason for your answer in (a).

